Same Motion, Different Scores: Learned Motion Evaluators Depend on Rotation Encoding

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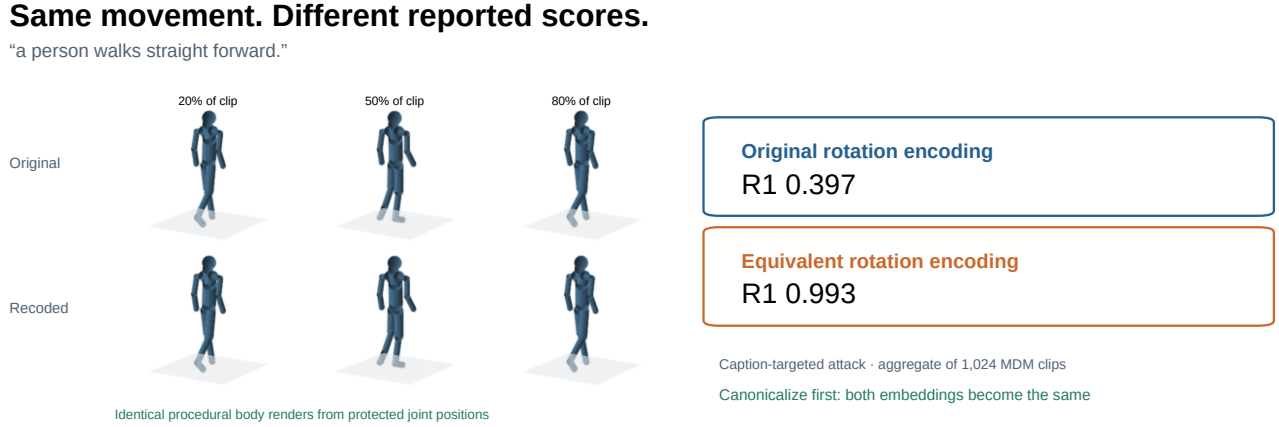


Figure 1. **The movement is identical; the score is not.** Both rows render the same joint positions and joint rotations of an MDM output; only the redundant coordinates of its 6D rotation features differ. A caption-targeted recoding of these coordinates raises the standard evaluator's top-1 R-Precision from 0.397 to 0.993 over 1,024 outputs. Canonicalizing rotations before evaluation gives both encodings the same embedding.

Abstract

A motion metric should score movement, not the numbers chosen to write it down. We show that widely used learned motion evaluators violate this requirement. The continuous 6D rotation representation admits positive rescalings and shears that leave every joint rotation unchanged; we apply them while holding positions, velocities, contacts and timing fixed. A fixed, caption-independent recoding raises the FID of real HumanML3D motion from 0.001 to 3.64, eight times the FID of MDM, and at equal input norm a fixed redundant direction moves FID 7.7× more than a fixed genuine rotation. Released outputs of all ten audited generator settings score differently once canonicalized, in six cases by more than the margin their authors report. With access to the evaluator, recoding alone drives retrieval to near-perfect accuracy in two evaluator architectures, lowers FID by up to 53% without captions, and raises sign-language backtranslation from 6.39 to 15.21 BLEU. Canonicalizing rotations before evaluation removes this dependence by construction while retaining all rotational information. We release an evaluation contract and a drop-in implementation.

1. Introduction

Generative motion models are compared by learned evaluators: a network embeds the generated motion, and retrieval accuracy or a Fréchet distance summarizes the embeddings [10, 26]. These scores decide which method is state of the art and, increasingly, which behavior a generator is fine-tuned toward [32, 33]. They rest on an assumption so basic that it is rarely stated: two descriptions of the same movement receive the same score.

This assumption fails. Figure 1 shows MDM outputs written in two ways. The rendered body, every joint position and every joint rotation are identical, yet the standard HumanML3D evaluator ranks the correct caption first for 99.3% of the recoded outputs instead of 39.7%. Nothing about the movement changed.

The cause is a well-known property of the continuous 6D rotation representation [43]. Two three-dimensional vectors

define a rotation through normalization and Gram–Schmidt orthogonalization, so scaling them by positive factors or adding a multiple of the first vector to the second leaves the rotation untouched. Forward kinematics and renderers discard these three redundant coordinates per joint. A learned evaluator that reads the raw vectors need not discard them, and we find that it does not.

Image generation has faced the same class of problem. FID changes with the resizing filter and compression applied to the same images [25], and it can be lowered without improving images by matching classifier statistics [15]; such findings changed how scores are computed and reported. Motion evaluation has no corresponding audit of its input representation. MARDM [21] showed that HumanML3D evaluators weight redundant feature groups heavily and proposed readers that discard them. We ask a sharper question: *can equivalent encodings of exactly the same rotations receive different scores?* Answering it requires interventions that change no represented quantity, and a remedy that keeps rotations, because some domains, such as the sign-language representations studied here, consist of rotations alone.

We answer the question at increasing levels of access to the evaluator:

- **No access.** A fixed recoding raises real-motion FID to eight times that of a published generator. At equal input norm, a fixed redundant direction moves FID 7.7× more than a fixed rotation of the joints.
- **Unmodified public outputs.** All ten audited release settings on HumanML3D and KIT-ML change FID when their rotations are canonicalized; in six, the change exceeds the 95% margin reported by their authors.
- **Evaluator gradients.** Recoding alone reaches 99–100% top-1 retrieval of the actual caption, and 79–97% for an arbitrary other caption, in both the T2M [10] and TMR [26] evaluators. Without any caption, it lowers FID by 39–53%, and the gain carries over to a second official checkpoint.
- **Rotation-only domains.** A layer that recodes generated sign-language motion raises backtranslation BLEU on PHOENIX-2014T from 6.39 to 15.21, and the failure reproduces on real How2Sign and CSL-Daily motion.

Throughout, the movement is verified to be unchanged, so none of these score changes can reflect better or worse motion, and no perceptual study is needed to interpret them.

The remedy is equally direct. Applying the converter's own canonicalization before feature normalization maps every equivalent encoding to one representative, so the score becomes a function of the represented movement. Unlike discarding rotations [21], it retains them and applies to existing evaluators without retraining. It is the identity on canonical real motion, and on noncanonical sign-language inputs, readers adapted to canonical features perform within 0.1 BLEU of raw-input readers.

Our contributions are:

1. an exact-equivalence audit of learned motion evaluators that isolates their dependence on rotation encoding, with physical calibration and equal-budget controls;
2. evidence that this dependence is present in public generator outputs and can be exploited without changing motion, across two evaluator architectures, two motion-language datasets and three sign-language corpora;
3. a canonical evaluation contract with a portable implementation that closes the channel while preserving rotational information.

2. Related work

Pitfalls of generative evaluation. Learned metrics are imperfect proxies for human judgment [14, 31, 37], and implementation details can move them as much as model changes: image resizing and compression alter FID [25], and FID can be lowered by matching class statistics rather than improving images [15]. We identify an analogous failure in motion evaluation that is purely representational. Because the evaluated movement is fixed exactly, the failure does not depend on how well the metric agrees with people.

Motion representations and evaluators. The 6D representation [43] and related rotation projections [16] are standard in motion learning. HumanML3D and KIT-ML features [10, 27] concatenate root motion, joint positions, velocities, 6D rotations and foot contacts, and are read by the T2M evaluator [10] and TMR [26]. Critics trained on human preferences [39] and reward fine-tuning of diffusion models [6] and motion generators [32, 33] make the consistency of such evaluators matter for training as well as reporting. MARDM [21] perturbs or discards redundant feature groups and trains readers on root motion and positions; its perturbations change represented rotations. We keep all represented quantities fixed and preserve rotations in the remedy.

Adversarial evaluation. Neural predictions can be redirected by universal or learned perturbations [1, 23, 29], including those of sign recognizers, image-quality models, translation metrics and text-to-motion systems [12, 17, 19, 22]. Our perturbations need not be small: they are exactly invisible to the represented motion. This allows a defense that is exact rather than empirical, unlike detectors, which adaptive attacks can evade [36].

Sign-language production. Production systems on PHOENIX-2014T, How2Sign and CSL-Daily [3, 9, 20, 42] are scored by backtranslation, in which a sign-to-text reader decodes the generated motion [4, 30, 38]. Recent work

adds semantic and linguistic assessment [7, 13]. We study a complementary requirement on the reader's input, in a representation that contains only body and hand rotations.

3. Evaluating movement, not coordinates

3.1. The rotation equivalence class

For one joint, let $d = (a, b) \in \mathbb{R}^6$ with $a \neq 0$ and b not parallel to a. The converter computes

$$u = a/\|a\|, \qquad b_\perp = b - (u^\top b)\, u,$$
$$v = b_\perp/\|b_\perp\|, \qquad R(d) = [u;\, v;\, u \times v]. \tag{1}$$

For $\alpha, \beta > 0$ and $\eta \in \mathbb{R}$, define the recoding

$$\mathcal{G}_{\alpha,\beta,\eta}(a, b) = (\alpha a,\ \beta b + \eta a). \tag{2}$$

The first normalized vector is still u, and orthogonalization turns the second into $\beta b_\perp$. With the canonical representative $\mathcal{C}(d) = (u, v)$,

$$R(\mathcal{G}(d)) = R(d), \qquad \mathcal{C}(\mathcal{G}(d)) = \mathcal{C}(d). \tag{3}$$

Conversely, any d with canonical representative (u, v) can be written $a = \rho u$, $b = \sigma v + \tau u$ with $\rho, \sigma > 0$. The recodings therefore span the entire fiber of encodings of a rotation, a three-dimensional set per joint and frame, rather than a neighborhood of similar rotations. This is a direct consequence of the converter [43]; our contribution is to audit evaluators against it and to close it.

We apply $\mathcal{G}$ only to valid rotation blocks (Fig. 2); every other channel and all padding is preserved bitwise. Rotation matrices, positions and forward kinematics are verified independently of any score (supplement). HumanML3D and KIT use the columns and the sign-language features the rows of R; each audit keeps the native convention.

3.2. Audits at increasing access

Fixed recodings use one global coefficient direction, $(\log\alpha, \log\beta, \eta) = (b, -b, b)$ with $b \in \{\pm 0.1, \pm 0.35\}$, and random coefficients as a control. They use no caption, generator or evaluator gradient.

Natural audits compare each unmodified public output with its own canonical representative, so the represented movement is identical by construction.

Targeted recodings optimize per-frame, per-joint coefficients, bounded as $(\log\alpha, \log\beta, \eta) = b \tanh q$, toward an actual or a shifted caption through the frozen evaluator.

Caption-blind recodings use no caption content and no test reference. With the evaluator's gradients, they move generated embeddings toward Gaussian-transport targets [2] computed from 2,048 training motions. Their outputs are frozen before any test-reference scoring, and a second official checkpoint that supplies no gradients also evaluates them.

Learned layers predict the same bounded coefficients from motion and the source sentence. Only the layer is trained, on training captions, with the generator and reader frozen. They are amortized attacks [1, 29] that need no per-clip optimization at test time.

3.3. Closing the channel

For a deterministic reader f, composing with canonicalization gives

$$(f \circ \mathcal{C})(\mathcal{G}(x)) = (f \circ \mathcal{C})(x) \tag{4}$$

for every admissible recoding, regardless of who chooses it or how. Detecting unusual encodings is weaker. In the supplement, a per-clip attack that preserves both the distance to the canonical encoding and the feature norm still raises PHOENIX backtranslation by 4.09 BLEU, while a threshold detector on that distance is at chance (AUC 0.50).

The order of operations matters. With feature normalization $\mathcal{N}$ and frame mask m, the canonical interface is $f(\mathcal{N}(\mathcal{C}(x)), m)$. Canonicalization must act on unnormalized features, because projecting standardized vectors does not implement the native converter, and the same interface should be applied to generated and reference motion. Figure 4 gives a guarded implementation that validates rotations, rejects degenerate vectors and preserves every non-rotation channel.

4. Experimental setup

Motion evaluators and data. HumanML3D has 263 features per frame, with 21 joint rotations in channels 67–192; KIT-ML has 251, with 20 rotations in channels 64–183 [10, 27]. The T2M evaluator [10] is a convolutional–recurrent motion encoder that reads all channels except foot contacts. We use its MDM-distributed weights and, as a replication, the distinct official weights released with T2M-GPT. TMR [26] is a Transformer with its own normalization that reads all 263 channels and scores by cosine similarity. For KIT-ML we use the native 251-feature T2M evaluator.

Public generators. We audit the released MDM [34], MLD [5], MotionLCM [8], T2M-GPT [40], T2M [10], MoMask [11], DiP [35] (which receives a 20-frame prefix) and MotionDiffuse [41] on HumanML3D, and MoMask and MotionDiffuse on KIT-ML, with their native inference settings. Each setting uses 20 native test draws of captions,

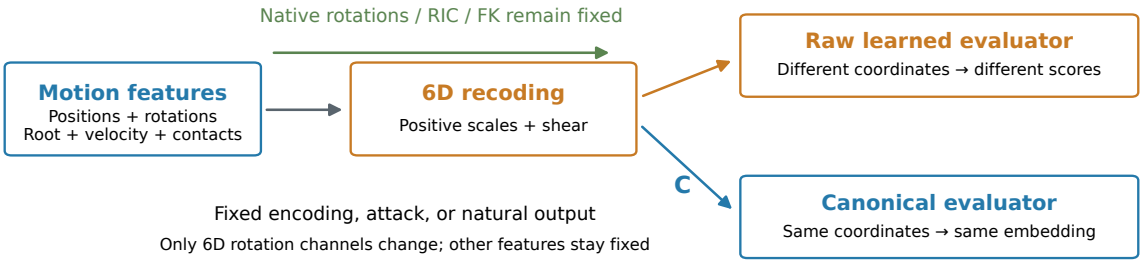


Figure 2. **Auditing and closing the channel.** A recoding changes only the redundant coordinates of the 6D rotation features; the converter returns the same rotations, and positions, velocities, contacts and timing are untouched. A raw learned evaluator sees different inputs for the same movement. Canonicalizing before normalization gives every equivalent encoding one input.

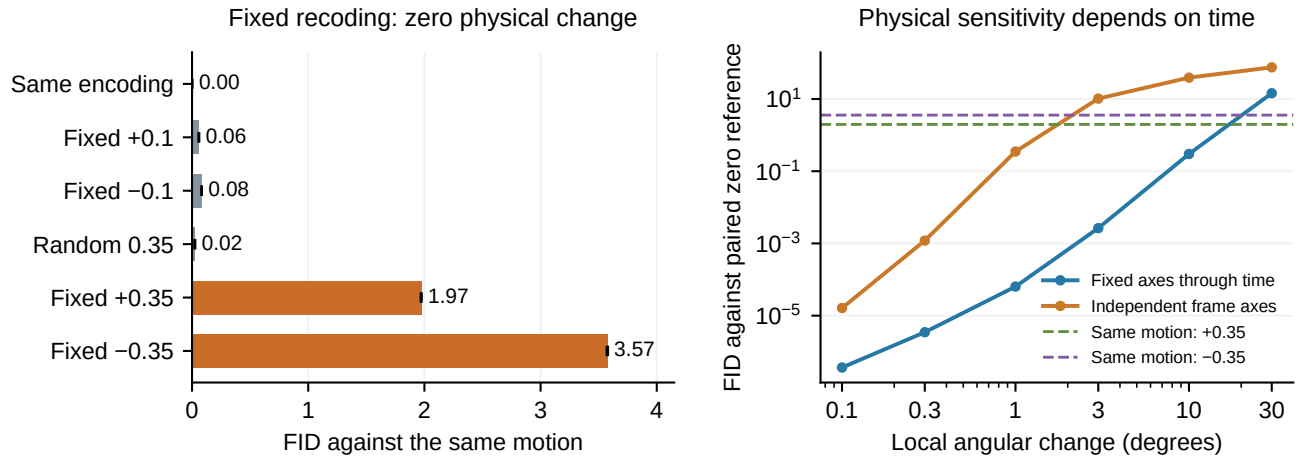


Figure 3. **A score change without a physical change, and its physical scale.** Left: fixed, caption-independent recodings of the same real HumanML3D motions, scored against those motions (no optimization or gradients; means and 95% intervals over 20 draws). Right: genuine rotation perturbations of the same motions, propagated coherently to all feature blocks, with an axis fixed through time or drawn independently per frame; dashed lines mark the two largest recodings. Both panels use the same reconstructed reference, so FID is zero by construction at zero change; the reconstruction itself has FID 0.02 to the original data.

```
import numpy as np
from tools.canonicalize_motion import (
    canonicalize_motion,
)
def reader_input(raw, lengths, mu, sd):
    x = canonicalize_motion(raw, lengths)
    m = np.arange(x.shape[1])[None, :]
    m = m < lengths[:, None]
    z = (x - mu) / sd
    return z * m[..., None], m
```

Figure 4. **A canonical evaluator input for HumanML3D/KIT.** Pass unnormalized features and the evaluator's own normalization statistics; use the same function for generated and reference motion.

crops and 32-candidate batches. Raw and canonical scores share outputs, captions and references, and paired t_{19} intervals are adjusted for multiple comparisons. These are consistency audits, not leaderboard reproductions.

Targeted and blind recodings. Targeted attacks use the same 1,024 test clips per source for both evaluators, 80 Adam steps and bound $b = 0.35$. Caption-blind attacks were run at two preset bounds; we report $b = 0.1$, on 960 MDM outputs for each of three generation seeds and on MLD, and give both bounds in the supplement. Equal-budget controls match the per-clip normalized change $\|D^{-1}(y - x)\|_2 = \kappa\sqrt{L}$, with the evaluator's standard deviations D and L valid frames.

Sign language. We use the released 240-feature body and hand rotation features and the public SignSparK back-translation readers [18, 20], one Sign Language Transformer [4] per corpus, on the full test sets of PHOENIX-

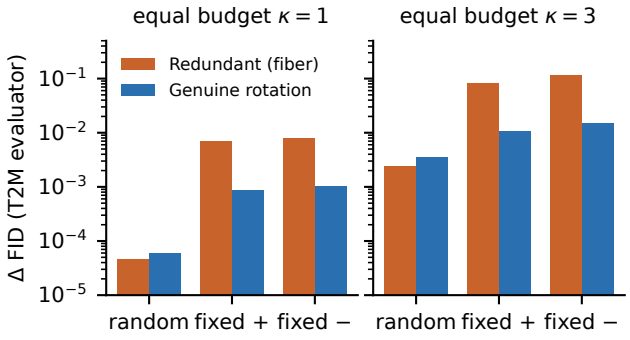


Figure 5. **Equal budget, unequal response.** FID change of real HumanML3D motion for perturbations of the same 126 rotation channels with equal normalized L_2 norm $\kappa\sqrt{L}$, mean over 20 draws. Redundant directions change no rotation; genuine directions rotate the joints. Intervals are narrower than the bars.

2014T (642 clips), How2Sign (2,183) and CSL-Daily (1,162). PHOENIX layers recode the outputs of a frozen, locally trained text-to-motion diffusion model; How2Sign and CSL-Daily layers recode real motion. Each layer is fitted with three seeds and evaluated with BLEU-4 [24, 28], word-level except on CSL-Daily, where it is character-level; paired bootstrap intervals are clustered by video on How2Sign.

5. Results

5.1. Equivalent encodings change scores

A fixed recoding is enough. Real HumanML3D motion has FID 0.001 against an independent real reference. After the fixed recodings with $b = +0.35$ and -0.35, the same movement has FID 2.00 and 3.64: 4.4× and 8.0× the FID of MDM's generated motion (0.455) under the same protocol. No caption, optimization or gradient is involved. Direction matters: magnitude 0.1 gives 0.06–0.08 and random coefficients 0.03. Figure 3 (left) shows the same pattern against a paired reference.

The scale is that of a physical change. The two large recodings move FID as much as genuine rotation perturbations of 1–3° drawn independently per frame, or of 10–30° about a fixed axis (Fig. 3, right). An encoding change that leaves the movement untouched is scored like a physical change of several degrees.

A fixed redundant direction outweighs a fixed rotation. At an equal normalized budget ($\kappa = 3$) within the same 126-dimensional rotation block, fixed recoding directions change FID by 0.082 and 0.117, against 0.011 and 0.015 for fixed directions that genuinely rotate the joints: 7.7× more response to coordinates that carry no movement, and 7.7–8.1× at the smaller budget (Fig. 5). The supplement reports all feature blocks and budgets.

Why. Real HumanML3D rotations are canonical, so the data on which these evaluators are trained never vary along a fiber. A recoding moves inputs off that distribution in directions the evaluator has never observed and was never trained to ignore. A fixed direction shifts every frame coherently and therefore moves the embedding statistics that FID compares, whereas random coefficients largely average out.

Table 1. **Public outputs receive different FID at identical movement.** $\Delta = \text{raw} - \mathcal{C}$ on the same outputs, 20 native test draws per setting; every 95% interval excludes zero. Published FID (± the authors' 95% margin) gives scale only, since protocols differ. †: $|\Delta|$ exceeds that margin. DiP is prefix-conditioned.

Release	Raw	$\mathcal{C}$	Δ	Published
HumanML3D				
MDM [34]	0.455	0.450	+0.005	0.489 ± .047
MLD [5]	0.445	0.449	−0.004	0.473 ± .013
MotionLCM [8]	0.575	0.606	−0.032†	0.467 ± .012
T2M-GPT [40]	0.183	0.149	+0.034†	0.141 ± .005
T2M [10]	1.393	1.347	+0.046†	1.087 ± .021
MoMask [11]	0.068	0.064	+0.003†	0.045 ± .002
DiP [35]	0.367	0.365	+0.002	0.228 ± .027
MotionDiffuse [41]	0.977	0.981	−0.004†	0.630 ± .001
KIT-ML				
MoMask	0.351	0.365	−0.014†	0.204 ± .011
MotionDiffuse	1.779	1.816	−0.036	1.954 ± .062

5.2. Public outputs are already affected

Generators do not produce exactly canonical rotations, so their reported scores already contain this dependence. Canonicalizing each output while keeping it otherwise identical changes FID in all ten settings (Tab. 1). The changes reach 0.046 and have both signs, including within diffusion models. They are comparable to differences that are reported as progress: in six of ten settings, the encoding-only change exceeds the 95% margin that the authors report for their own FID. The mean ranking of the five original releases is unchanged, and the corresponding retrieval and TMR gaps are small (supplement). On public outputs, the effect is a measurable inconsistency rather than a reordering of the field.

5.3. Recoding alone can redirect evaluators

Retrieval reaches its ceiling. With evaluator gradients, recoding raises top-1 R-Precision from 0.397 to 0.993 for MDM outputs and from 0.462 to 0.996 for real motion with the T2M evaluator, and from 0.578 to 0.993 and from 0.688 to 1.000 with TMR (Tab. 2). The evaluators are not recognizing better movement: when the target is another clip's caption, retrieval of that wrong caption rises to 0.79–0.97 while the actual caption is lost. Attacks optimized through a canonicalizing evaluator do not move scores, and canonical embeddings change by at most 6.7×10^{-6}.

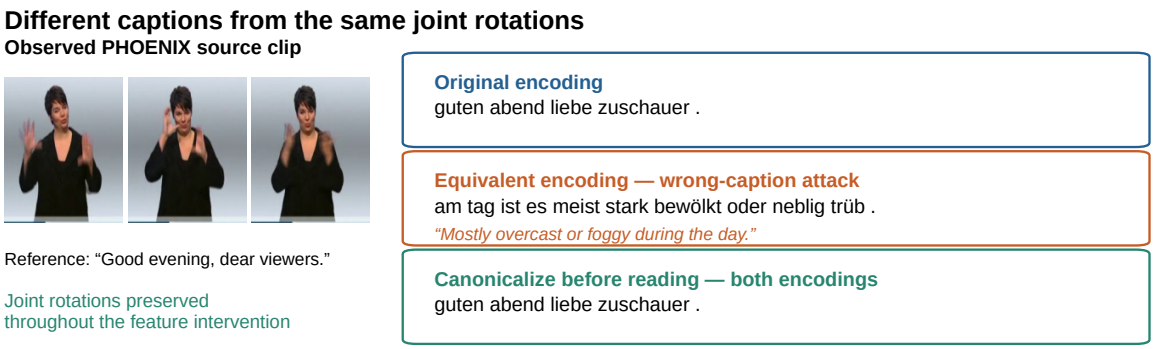


Figure 6. **A greeting becomes a weather forecast.** A real PHOENIX-2014T test clip (RGB frames shown for reference) and its released rotation features. A per-clip recoding toward another test sentence leaves every joint rotation unchanged, yet the backtranslation reader decodes the weather report; canonicalizing restores the original decoding for both encodings. The clip was chosen for illustration; aggregate results are in Tab. 3.

Table 2. **Retrieval follows the encoding, not the movement.** R-Precision (top-1) on the same 1,024 test clips per source, before and after a caption-targeted recoding (80 steps, bound 0.35). *Shifted*: the target is another clip's caption and R1 is measured against it; its *before* value is the canonical-gradient control, which leaves the input unchanged. Canonicalization maps attacked and unattacked inputs to the same scores in every row.

		Actual caption		Shifted caption	
Evaluator	Motion	Before	After	Before	After
T2M [10]	MDM	0.397	0.993	0.033	0.788
T2M [10]	Real	0.462	0.996	0.028	0.839
TMR [26]	MDM	0.578	0.993	0.017	0.946
TMR [26]	Real	0.688	1.000	0.010	0.968

Captions are not needed to lower FID. Using only statistics of training motion, caption-blind recoding lowers MDM FID from 0.384 to 0.210, from 0.376 to 0.176 and from 0.327 to 0.199 for three generation seeds (39–53%), and MLD FID from 0.706 to 0.289. Scored by the second official T2M checkpoint, which supplied no gradients, the same frozen outputs still improve by 12–25% (supplement). These reductions exceed the FID differences between several published generators (Tab. 1) and are obtained without moving a single joint.

5.4. Rotation-only domains: sign language

Sign-language features contain only rotations, so discarding them is not an option. A learned layer that recodes the PHOENIX-2014T generator's outputs from their source sentence raises backtranslation BLEU from 6.39 to 15.21, comparable to the 15.00 of real signing, while every canonical hypothesis is unchanged (Tab. 3). Changing only the sentence given to the layer moves the decoded output toward that sentence: BLEU against it rises from 0.83 to 9.60.

Table 3. **Backtranslation changes while the signing does not.** BLEU-4 on full test sets for a learned recoding layer (mean of three fitting seeds; paired 95% interval). The canonical reader's hypotheses are identical before and after the layer. CSL-Daily BLEU is character-level.

Corpus	Motion	Raw	+Layer	$\mathcal{C}$	Gain [95% CI]
PHOENIX	Generated	6.39	15.21	6.39	8.81 [7.82, 9.87]
How2Sign	Real	3.48	6.11	3.48	2.64 [2.27, 3.03]
CSL-Daily	Real	6.89	7.43	7.07	0.54 [0.14, 0.97]

On real How2Sign and CSL-Daily motion, the same construction raises BLEU by 2.64 and 0.54, with intervals excluding zero. Per-clip optimization exposes a larger channel still: on PHOENIX, the reader decodes an arbitrary other sentence at 85.19 BLEU (Fig. 6 and supplement).

5.5. Canonicalization closes the channel

Exact closure. Canonicalized inputs give identical embeddings and identical decoded hypotheses for every targeted, caption-blind and learned recoding above, as Eq. (4) predicts; numerical checks bound the residuals (supplement). Readers that discard rotations [21] are also invariant, but they replace the evaluator and cannot apply to rotation-only inputs. Canonicalization keeps the evaluator, and its ranking of the eight HumanML3D releases agrees with that of the public essential-feature readers (Kendall $\tau = 0.857$).

No measured accuracy cost. Real HumanML3D and KIT rotations are canonical, so real-motion retrieval is unchanged by construction. Real sign-language features are not. Readers fine-tuned on canonical inputs reach How2Sign and CSL-Daily test BLEU within $+0.10$ and -0.02 of matched raw-input readers, with intervals containing zero.

6. Discussion

A consistency checklist. Motion scores should be specified on represented movement. For any evaluator of rotation features we recommend four steps: (i) name the converter and its canonical representative; (ii) report each method's raw-minus-canonical score gap, since a nonzero gap shows that the evaluator reads the encoding; (iii) score generated and reference motion through the same interface $f(\mathcal{N}(\mathcal{C}(x)), m)$ (Fig. 4); and (iv) canonicalize learned rewards before their gradients reach a generator, because the redundant coordinates are exactly the directions a generator can change at no cost to its motion.

Beyond 6D. The audit relies only on a converter that maps many encodings to one rotation, so it applies unchanged to other parameterizations. Unnormalized quaternions are invariant to positive scale and to sign, 9D matrices projected by SVD [16] to their symmetric factor, and axis–angle vectors to full turns. In each case the test is the same: recode within the converter's fibers, compare scores, and canonicalize before reading.

Scope. Our claims concern evaluation. We do not claim that published results were manipulated or that standard training exploits the channel. Canonicalization removes rotation-encoding dependence; it does not make an evaluator accurate, and it does not address redundancy between positions and velocities or attacks that change the movement. The sign-language readers come from one released family, and their absolute accuracy on How2Sign and CSL-Daily is low.

7. Conclusion

Learned motion evaluators assign different scores to the same movement when its rotations are written differently. The dependence is present in released generator outputs, can be as large as a physical change of several degrees, and can drive retrieval, FID and backtranslation without moving a joint. Canonicalizing rotations before evaluation removes it exactly and keeps all rotational information. As happened for image metrics, we hope that a precise statement of the evaluation interface becomes part of how motion results are computed and reported.

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